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Leveraging AI for Inventory Management and Accurate Forecast – An Industrial Field Study

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Abstract

Accurately forecasting demand is one of the most undervalued and complex strategies that can significantly impact organizations bottom line. This industrial field study was co-conducted with Sumitomo Corporation's Tubular Division which primarily deals with high-grade Oil Country Tubular Goods (OCTG) globally. The presented solution demonstrates how with the right data set (drilling sequence data, stock data and consumption data), artificial intelligence can be used to build out a model that can quantify and predict future demand accurately thereby reducing cost, working capital and emissions.

Multiple multi-layered machine learning models were built to compare and analyze a wide variety of data inputs for bill of materials, operational/project schedules; This includes (a) 'product movement data' which describes the changes in demand and supply of a product, (b) 'product specification data' which describes the characteristics of a product, and (c) 'activity specification data' which describes the characteristics of an activity. The models follow the base temporal map design with different weighting on model inputs. With a temporal map, a sequence of monthly data values (called lags) is used to predict the next monthly value in the sequence. The lags are rolled so that there are six months of data for the model to predict on. All models also use boosted decision-tree-based ensemble machine learning algorithm.

It is critical to understand how product movement metrics (actual and safety stock levels, historical forecasts, and consumption patterns), product specification data (lead time, product grade, well function, well category, work center), and external factors (oil price, rig counts, national budget, production targets) can be utilized together to better understand future product demand. Using historical data acquired from drilling operations and supply chain over an eight-year period, multiple machine learning models were trained to predict one year of demand across the most consumed products. Across five years of predictions (2016 to 2019), the models were able to predict with 78% average accuracy for the top 10 products by volume which represents 75% of inventory volume. Across the same time-period, they were able to predict with 73% average accuracy on all 17 products which account for 80% percent of inventory volume. Further iterative updates with additional data led to improvement in results and the model where the model predicted with an improved accuracy of 83% on the top 17 products and an accuracy of 86% on the top 10 products.

Moreover, the data can also be used to generate dashboards featuring metrics on material uncertainty / velocity and expected differences between the internally predicted forecasts and actual sales. The results

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further indicate that, on average, and within a simulated environment (where shipping delays were not considered for instance,) the AI model can maintain a lower inventory than the originally planned stock levels at lowest cost and footprint. This would not only lead to less resource consumption, but also reduce the embodied carbon and emissions within the overall process.

This novel study presents the success of a validated tailored AI model for inventory forecast with field data and commercial implementation. Such a tool can be integrated into other value adding digital tools, such as integrated schedule optimization, logistics optimization and management systems to make overall operations more efficient and sustainable with lower costs, inventory, wastage, and reduced emissions.

Keywords: Artificial Intelligence, Optimization, Economics, Inventory Management, Emissions

Introduction

With the increasing energy demand, geopolitical constraints, and volatile market conditions, supply chain operations in the hydrocarbon industry have become more complex over the past few years. As we head towards an era of greater uncertainty associated with increasing risk, competitiveness, and scarcity of resources; accurately forecasting demand is one of the most undervalued and complex strategies that can significantly impact organizations bottom line.

The success of an exemplary inventory management solutions often relies on an organization's ability, capability, and capacity to streamline and integrate the complete end to end process effectively. This may include manufacturing, sourcing, processing, transportation, and end-delivery to the clients. With the advent of intelligent solutions, organizations have realized the importance of take advantage of real-time data to make informed decisions within supply chain that can significantly affect the company's bottom line. One such implementation is through the application of artificial intelligence (AI) and machine learning solutions that can assist organizations to make complex decision in real-time through a data centric approach.

Over the years, there has been numerous promising applications through AI particularly for oil and gas applications [1–10]. Even so, there are considerable gas and multiple domains where there is significant potential to enhance current processes in place as illustrated in Figure 1. These include challenges related to involving inventory management, purchasing, location planning, freight consolidation, and routing/scheduling problems. This industrial field study was co-conducted with one of the dominant global suppliers, Sumitomo Corporation's Tubular Division, which focuses on supplying a wide variety of tubular products including high-grade Oil Country Tubular Goods (OCTG), OCTG accessories and line pipe to companies in the oil and gas industries. In addition to their expertise in their proprietary supply chain management system which provides comprehensive well management services, they aim to expand the value chain beyond traditional services including demand forecasting, as it is one of the most undervalued strategies for increasing profitability. Accurate forecasting benefits businesses in many ways that ultimately increase the bottom line. Therefore, it is critical to understand how product movement metrics (actual and safety stock levels, historical forecasts, and consumption patterns), product specification data (lead time, product grade, well function, well category, work center), and external factors (oil price, rig counts, national budget, production targets) can be used together to better understand future product demand. Such a tool can be integrated into other value adding digital tools, such as integrated schedule optimization, fleet and route optimization and connected Driller/worker. The industrial field study presented within the manuscript demonstrates how with the right data set (drilling sequence data, stock data and consumption data), artificial intelligence can be used to build out a model that can quantify and predict future demand accurately while reducing the associated emissions. Hence, the major objectives of this paper are to:

a. Demonstrate how AI can be used to better quantify and predict future demand accurately. This includes using historical data effectively to improve accuracy of future orders in place along with reducing lead times.

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b. Leverage AI to identify emerging issues that might be unseen by the organization.
c. Integrate advanced analytics to enhance decision making process in run-time.
d. Improve transparency and trust for all stakeholders through one system while reducing human error.

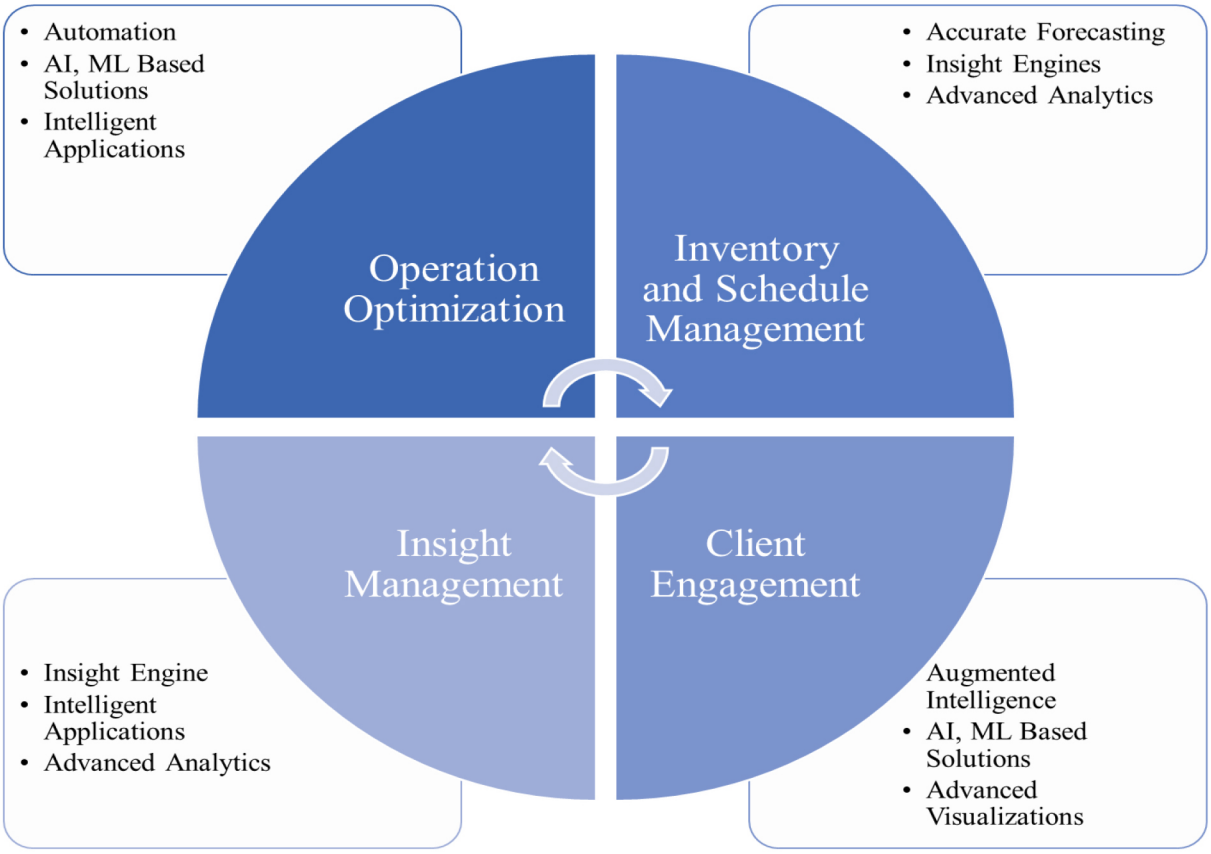

Figure 1—Potential Supply Chain Domains for AI Augmented Applications

Methodology

For this industrial case study, multiple data sets were screened and analyzed. Among them, the key sets of data included

a. Historical drilling sequence forecasts and basket (products required for each Well type) data
b. Historical stock for each product, and
c. Historical consumption for each product

Drilling sequence and basket data was used to create a 'forecasted quantity' that is the projected demand of a particular product at a future month. The sequence is updated every month and new projections are created. Stock data specified the amount of unrestricted stock available at any given time for use in a well. Consumption data primarily indicates the quantity shipped per product along with date and destination. This data was used to construct historical consumption patterns for each product. Once a comprehensive analysis was conducted, multiple multi-layered machine learning models were built to compare and analyze a wide variety of data inputs; primarily

a. 'product movement data' which describes the changes in demand and supply of a product,
b. 'product specification data' which describes the characteristics of a product, and
c. 'activity specification data' which describes the characteristics of an activity.

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Consumption Data Analysis

Each product has a historical consumption pattern from the data provided. This allows to track changes in consumption patterns across time in terms of products consumed by operator. The graph below in Figure 2 shows the consumption patterns for three products (1, 2, and 3).

There are three distinct consumption patterns: high volume movement in product 1, mid-level movement in product 2, and slow product movement with 3. This analysis was used to create a separation of products into fast, mid, and slow movers that allowed to predict product consumption more accurately.

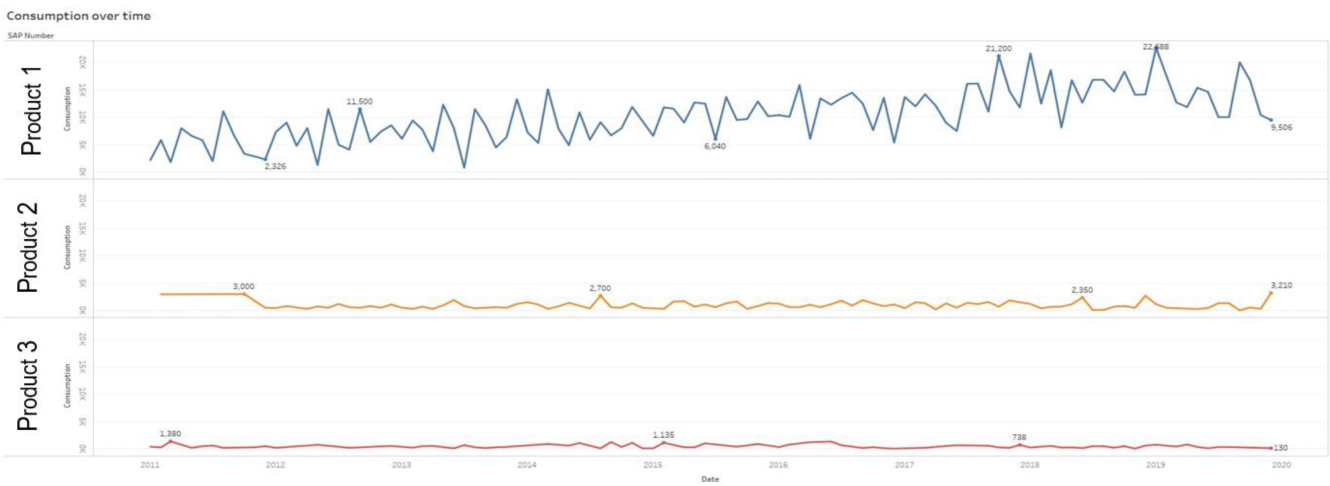

Figure 2—Example of products falling into the fast, mid, and slow mover categories

Stock Data Analysis

Furthermore, from the data it was evident that each product's stock level varies based on the product's demand. Figure 3 further highlights how the three products (1, 2, and 3) that have distinctly different stock level patterns across the years. The sections with flat or zero-level stock, such as a brief period during early to mid-2019 for product 3, may signify that the product is either out entirely out of stock and that there might be a shortage at the yards; or that the part is no longer in rotation as a part that is being ordered. Analyzing these patterns can assist to shape the data to assist the model better.

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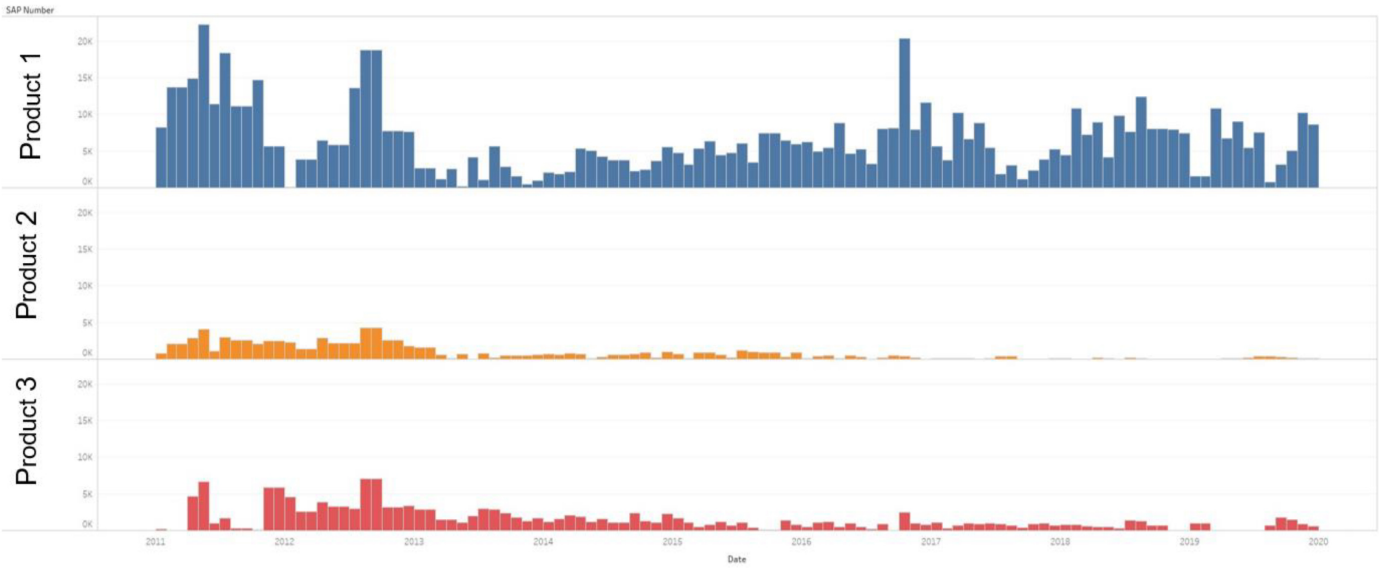

Figure 3—Example of products stock data to analyze demand

Drilling Sequence Data Analysis

The drilling sequence was analyzed to identify relationships between the projected demand vs. actual demand. To quantify and extract value from these relationships, the concept of a 'drilling sequence path' was created. A drilling sequence path is a path that projected drilling activities take through the drilling forecasts from when they first appeared on the drilling sequence until they are drilled. These paths capture the delays and movement of drilling activities as each new iteration of the drilling sequence is published. These paths allow to answer two important questions; how certain that a well is likely to be drilled, and when a well is likely to be drilled.

Sparsity Analysis

Sparsity analysis is performed to understand variations in the use of the product and to identify any missing data. The sparkline created from this data as shown in Figure 4 summarizes the general shape of the data completeness and points out the rows with the maximum and minimum consumption across time. Missing consumption refers to no consumption entirely for a single or all products for any period greater than one month.

From the detailed analysis, it was observed that the missing data has a significant impact on the model, as the model uses historical consumption patterns as inputs. Missing consumption data may have resulted from contract renegotiations, breaks in delivery, or even data quality issues. These gaps were considered during the machine learning model design. This type of analysis can also be used in future phases to provide quality assurance of forecast data coupled with drilling data.

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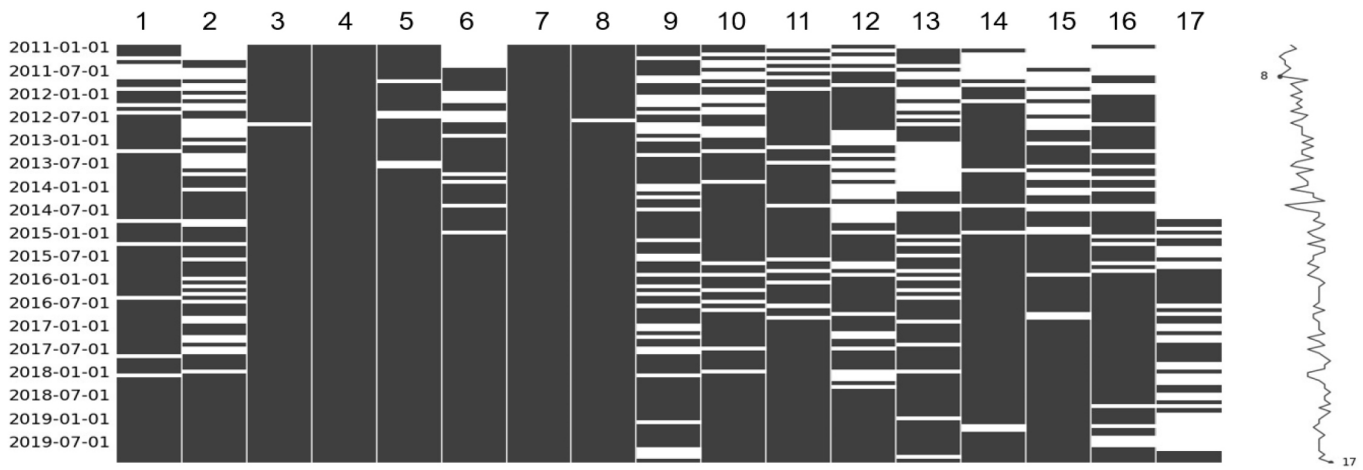

Figure 4—Sparsity Analysis

Basket Data Analysis

Basket data contains the predefined requirements lists for a specific well design. The baskets follow iteration cycles that receive updates periodically. These baskets are linked to the forecasts and give insights into what products are needed for each well that is to be drilled. Missing basket data does not significantly impact the model as it is to be noted that that baskets can be older and not have received updates for periods greater than three months. Like the other datasets, this analysis was taken into consideration while forming the model design

The models follow the base temporal map design with different weighting on model inputs. With a temporal map, a sequence of monthly data values (called lags) is used to predict the next monthly value in the sequence. The lags are rolled so that there are six months of data for the model to predict on. All models also use boosted decision-tree-based ensemble machine learning algorithm. A high-level simplistic illustration of the entire flow process is depicted in Figure 5.

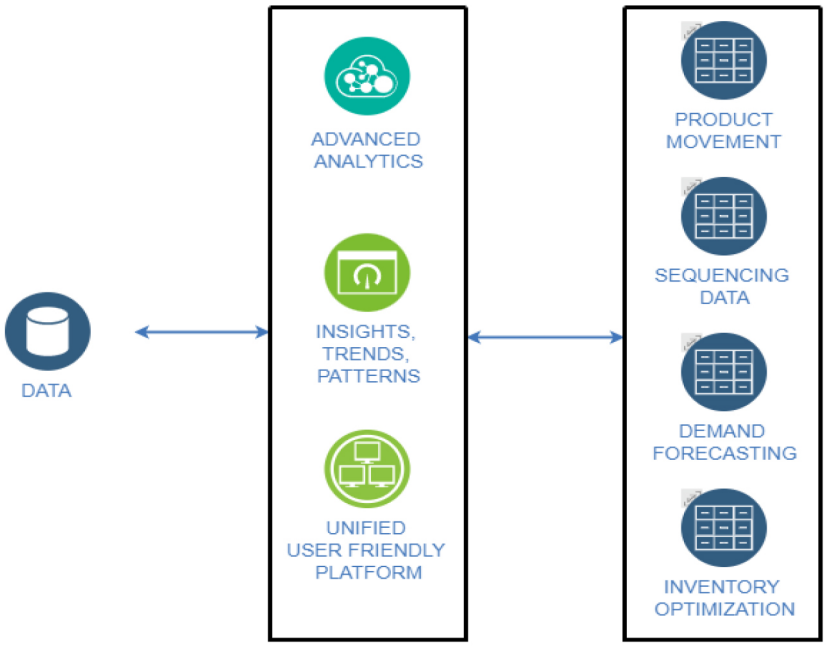

Figure 5—Simplistic Illustration of Workflow

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Results and Discussion

For the model development, an agile approach was taken. Each iteration cycle varied in length depending on the learnings of each development cycle. The result was three cycles, where cycles one and two were preliminary cycles to understand better how the model could be formed, how the model interprets the data, and what our performance potential was. Each cycle followed a minimum of a 3–4-week development cycle. When sufficient learnings or performance were achieved, an enhanced approach would improve performance and accuracy, or closer align the model output with the business case and requirements. The models were tested on the Top 17 products, which approximately represent 80% of the total product volume consumed for the industrial application.

The tables each contain the following information: Data trained on, Year Predicted and Results.

a. 'Data trained on' refers to the data that the model has available to learn from during that cycle. For instance, if data trained on is 2011 – 2014, that means the model had the data (drilling sequence data, stock data and consumption data) from 2011 through 2014 to learn from.
b. 'Year predicted' refers to the year that the model was tested on before seeing the actual results. For instance, if year predicted is 2015, then the model was used to predict 2015 results before those results were released. This can ensure that the model did not learn anything about the 2015 results before the 2015 predictions were made.
c. Results are given in terms of Mean Absolute Percent Error or MAPE, a common way of reporting results with aggregated percent errors. MAPE forces the use of an absolute value for the percent error when averaging across multiple percent errors. This ensures that positive and negative errors do not cancel each other out and gives a much more accurate average percent error value. For instance, if a MAPE is reported as 45% across the top 17 products, this means that the percent error was calculated for each product, each percent error was made positive (a -15% error become a +15% error so that the -15 and +15 percent errors did not cancel each other out in the calculation) and finally the mean of these errors was calculated to arrive at the final answer. The formula for MAPE is given below:

$$M = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{A_t - F_t}{A_t}\right|$$

M = mean absolute percentage error
n = number of times the summation iteration happens
A_t = actual Value
F_t = forecast Value

Accuracy is then calculated as 1-MAPE.

The final models were evaluated for accuracy and performance in predicting 2016, 2017, 2018, and 2019 total consumption and in predicting total consumption grouped into slow, medium, and fast movers. The two tables below show the results.

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Table 1—Actual vs Predicted consumption for all models across 2016-2019

Training data	Forecast year	Model 1	Model 2	Model 3	Actuals	MAPE	Accuracy
2011-2015	2016	363518	320080	328868	375833	10%	90%
2011-2016	2017	376221	341004	354356	391422	9%	91%
2011-2017	2018	350568	379840	390086	380802	2%	98%
2011-2018	2019	430904	418539	425613	366828	16%	84%
	Total	1521211	1459463	1498923	1514885	2%	98%

The table above is the aggregated yearly consumption for the top 17 products. It is seen that the model(s) does an excellent job of predicting the amount of material to be used. In fact, over four years, 1,514,885 units of material was consumed, and the average prediction across all three models is 1,493,199 units.

Table 2—Actual vs Predicted consumption for all models and product categories across 2016-2019

Training	Forecast	Group	Model 1	Model 2	Model 3	Actuals	MAPE	Accuracy
2011-2015	2016	Slow	25352	29494	26521	34231	21%	79%
		Medium	40342	32352	39564	43722	14%	86%
		Fast	297824	258234	262783	297880	8%	92%
2011-2016	2017	Slow	21661	28015	23725	33073	26%	74%
		Medium	54641	39936	46149	52601	13%	87%
		Fast	299919	273053	284482	305748	7%	93%
2011-2017	2018	Slow	25464	29416	22687	26206	10%	90%
		Medium	52800	42838	49385	44646	11%	89%
		Fast	272305	307587	318014	309950	5%	95%
2011-2018	2019	Slow	26440	32695	28746	24931	17%	83%
		Medium	54757	41369	47113	42036	15%	85%
		Fast	349707	344476	349753	299861	16%	84%

Table 2 shows the aggregated yearly consumption for the same 17 products but grouped into slow, medium, and fast movers. The final column (MAPE) shows the mean of the absolute error for each row. Upon further breakdown of the data across five years of predictions (2016 to 2019), the models were able to predict with 78% average accuracy for the top 10 products by volume which represents 75% of inventory volume. Across the same time-period, they were able to predict with 73% average accuracy on all 17 products which account for 80% percent of inventory volume. Further iterative updates with additional data led to improvement in results and the model where the model predicted with an improved accuracy of 83% on the top 17 products and an accuracy of 86% on the top 10 products.

Impact

The AI solution has impact in three areas:

a. The first area of impact is the ability to challenge current forecast schedule. The key here is understanding what the error of the AI model means and how that error compares to the error in the traditional forecast method.

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b. The second area of impact is on the inventory. To demonstrate this, a simple inventory simulation that allows comparison between a hypothetical AI Inventory and the actual reported inventory levels was created.
c. The insights can be coupled with the drilling sequence data for optimal prediction and recommendations. The right data set can be used to leverage OCTG supply chain to build scenarios for drilling schedule with minimal lead time.

The model error directly in the simulation which also helps to understand what 70% or 80% accuracy really means in terms of expected impact. If the range of the traditional forecast is greater than the model error, then this means that on average, the AI model will be more accurate than the traditional forecast.

For example, for Product 3, results indicate that the difference (or error) between the AI forecast and the actual consumption is about 270 units on average (accounting for lead time). The average difference between the traditional forecast and actual consumption is around 700 units. What this really means is that the AI forecast can be used to identify, challenge, and correct the traditional forecast when the two forecasts differ by more than 270 units. This, in turn, means that the large "misses" in the traditional forecast can be challenged and possibly corrected before the impact is felt on inventory levels or in other areas.

Figure 6 highlights the average difference between the AI forecast and the traditional forecast. The lower error range (~270 units vs ~700 units) of the AI forecast allows to identify, challenge, and correct large "misses" in the traditional forecast. For instance, if the AI forecast is predicting 1200 and the traditional forecast is predicting 1600, then the prediction of 1600 can be challenged on the basis that it is outside the range of the AI forecast (prediction plus/minus average error). The key outcome is that the AI model provides a way to identify where the traditional forecast might be off or misleading and provides the necessary information to challenge it along with specifying how to correct it.

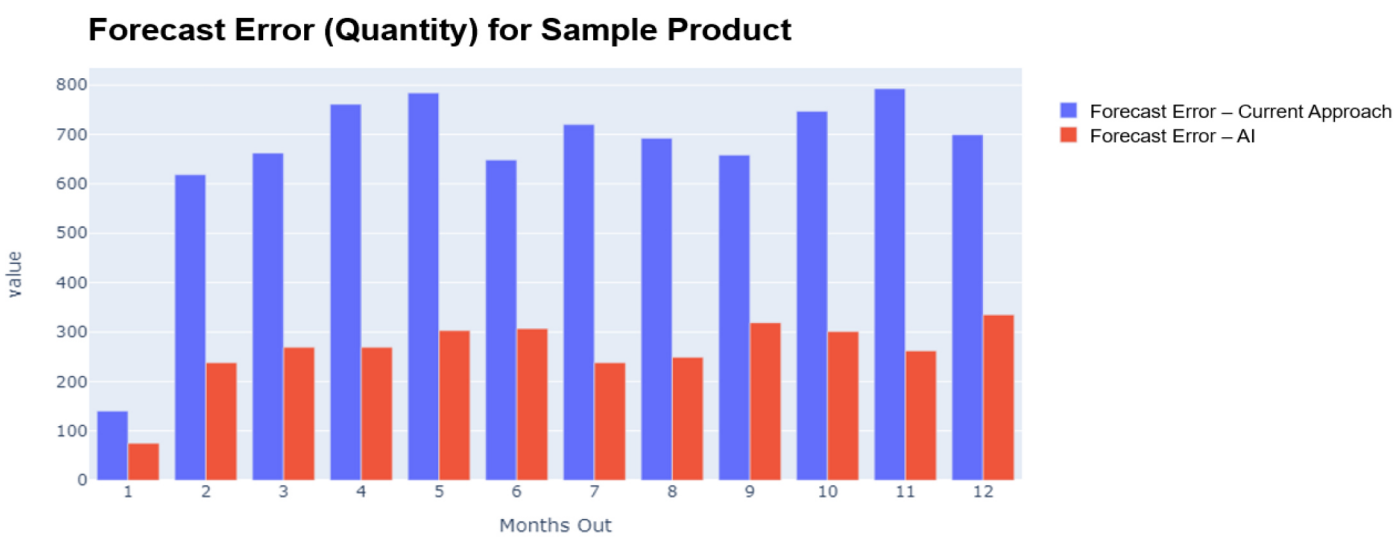

Figure 6—Difference in Forecast between Traditional and AI Approach

Furthermore, a simple inventory simulation was performed from 2019 onward by using the average error of the model to simulate what the AI predictions would have been if trying to forecast from 2019. These predictions were then used to "place an order" and inventory was calculated with the usual formula of prediction/order – actual consumption + current inventory and tracked on a cumulative basis. The results assist to visualize current accuracy means in terms of impact on inventory levels and helps understand how that could improve in the future as the model becomes more accurate with additional data. The simulation was run on three products (top three products by volume which also cover a range of prediction accuracy). The results generated indicated that, on average, and within a simulated environment (shipping delays where

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not considered for instance,) that the model can maintain a lower inventory than the current stock levels. This would not only lead to less resource consumption, but also reduce the embodied carbon and emissions within the overall process. Moreover, the data can also be used to generate dashboards featuring metrics on rig uncertainty / velocity and expected differences between the internally predicted forecasts and sales.

A simplistic case study was conducted in-house to track emissions and to understand the true impact in reduction though optimization of materials. For this simplistic study, there are numerous assumptions. This includes selection of a particular manufacturer for which data was readily available and where reasonably accurate calculations can be made to track the emissions during the entire journey till site. A more robust strategy that accounted for these assumptions would need to be implemented before the simulation results could be replicated with certainty. The main aim is to demonstrate expected future impact once model is implemented. Table 3 illustrates the key metrics for emissions from source to well for 1 ton of steel for a sample high moving material. The calculation comparison was extended to same material which is also manufactured in another country, Japan as compared to China. It is interesting to note that for the same material, even though the transportation distance is significantly larger in the case of Japan to the destination well site, the overall emissions is less. The primary reason for this includes the source of energy mix used for manufacture for materials in both countries along with the distance between manufacturer to the port within the origin country. Hence, for this study, for each ton of inventory level reduction, we have reduced the associated emissions by around 1 ton of CO2e. This clearly indicates the complexities, challenges, and potential opportunities when it comes to true emission tracking while also evaluating the true impact of decisions and optimizations.

Table 3—Emissions for 1 ton of sample steel material from source to well based on country of origin – China and Japan

Emission Type	Emissions - China (Kg CO2e per ton of material)	Emissions - Japan (Kg CO2e per ton of material)
Embodied	728.05	655.25
Transportation (Origin -> Destination) (Shipment)	97.6	114.37
Local Transportation (Origin Country)	86.8	77.3
Local Transportation (Destination Country)	49.6	49.6
Storage & Transhipment (Origin)	43.2	54.2
Storage & Transhipment (Destination)	44	44
Total (Kg CO2e per ton of material)	**1049.25**	**994.72**

Conclusion

Accurately forecasting demand is one of the most undervalued and complex strategies that can significantly impact organizations bottom line. This industrial field study co-conducted with a dominant player in the industry, demonstrates how with the right data set, AI can be used to build out a model that can quantify and predict future demand accurately while reducing the associated emissions. Key conclusions are as follows:

- Using historical data acquired from field operations over an eight-year period, multiple multi-layered machine learning models were trained to predict one year of demand across the most consumed products.

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- Across five years of predictions (2016 to 2019), the models were able to predict with 78% average accuracy for the top 10 products by volume which represents 75% of inventory volume. Across the same time-period, they were able to predict with 73% average accuracy on all 17 products which account for 80% percent of inventory volume.
- Further iterative updates with additional data led to improvement in results and the model where the model predicted with an improved accuracy of 83% on the top 17 products and an accuracy of 86% on the top 10 products.
- AI model provides a way to identify where the traditional forecast might be off or misleading and provides the necessary information to challenge it along with specifying how to correct it.
- Furthermore, the AI model within a simulated environment (shipping delays where not considered for instance,) shows potential to maintain a lower inventory than the current stock levels.

The results generated from this investigation shows considerable promise for accurate forecasting demand with the right data sets and techniques. Further iterations of the model aim to expand the learnings to optimizations within logistics, drilling sequence, end to end emission tracking within supply chain and enhancing HSE through 'Smart Alerts' to make informed business decisions while making operations more sustainable.

Acknowledgement

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